

***Translator's note.** This document is an unofficial English translation, provided for the reader's convenience. The original letter of recommendation was written and signed in French, and the French version is the only authoritative one. In case of any discrepancy, the original French text prevails.*

Subject: Letter of recommendation for Mr. Guillaume Guguen

Dear Sir or Madam,

It is with great pleasure that I write to recommend Mr. Guillaume Guguen, who completed his internship within the Direction Générale de l'Armement (DGA) under my direct supervision from 01/02/2026 to 31/07/2026.

During this period, Guillaume joined a team designing a counter-drone interceptor drone, his main task being the integration of guidance laws that took into account the shape of the vehicle (a quadrotor) and the detection of a target by means of a camera.

Guillaume demonstrated great rigour in the structure and readability of the software development, and a remarkable intellectual capacity to reach his objectives (such as the custom development of an EKF to obtain high-performance target-interception laws). The quality of his work could have been the subject of a conference paper. He was also able to use the same simulator to integrate, at my request, collaborative guidance laws. This code was adapted to run on several commercial drones and was able to fly drones under my responsibility in a flight arena. While the formation-flight part could not be completed for lack of time, the "trajectory-tracking" part with a single drone was validated, with close results between simulation and real flight.

Among his professional qualities, I would like to highlight:

- **His rigour:** Guillaume developed his code with rigour and clarity, testing it through numerous simulations to tune various parameters and validate results. He also met the deadlines set in a demanding environment.
- **His adaptability:** He was able to adapt to his working environment (the vehicle, the camera and the simulation framework were imposed on him), draw on his knowledge and carry out a state of the art that led him to solutions for intercepting a drone (an EKF coupled with a kinetic interception guidance law). He handled drones successfully and adapted to the hardware constraints.
- **His integration into a team:** Guillaume fitted harmoniously into our team, showing attentiveness, loyalty and an excellent team spirit—qualities that are essential within the DGA.

I am convinced that Guillaume will be a valuable asset to your organisation. He has the intellectual and human foundations needed to succeed in complex and demanding environments. I remain fully at your disposal for any further information regarding his internship.

Please accept, Dear Sir or Madam, the expression of my highest regards.

Charles Coquet

Guidance/Control Engineer specialising in autonomous agents, DGA-MI

Done in Bruz, on 21/07/2026